

Why Did Operational Throughput Drop Last W

A Data-Driven Investigation into Mid-Year Flow Rate Fluctuations

Targeting Logistics, Weather, and Governance across Nairobi, Mombasa & Eldoret Terminals

THE OPERATIONAL PROBLEM

During the week of June 25 to July 1, 2026, the central operations console flagged a significant reduction in mean flow rates (LPM) at several key depots, threatening daily distribution targets.

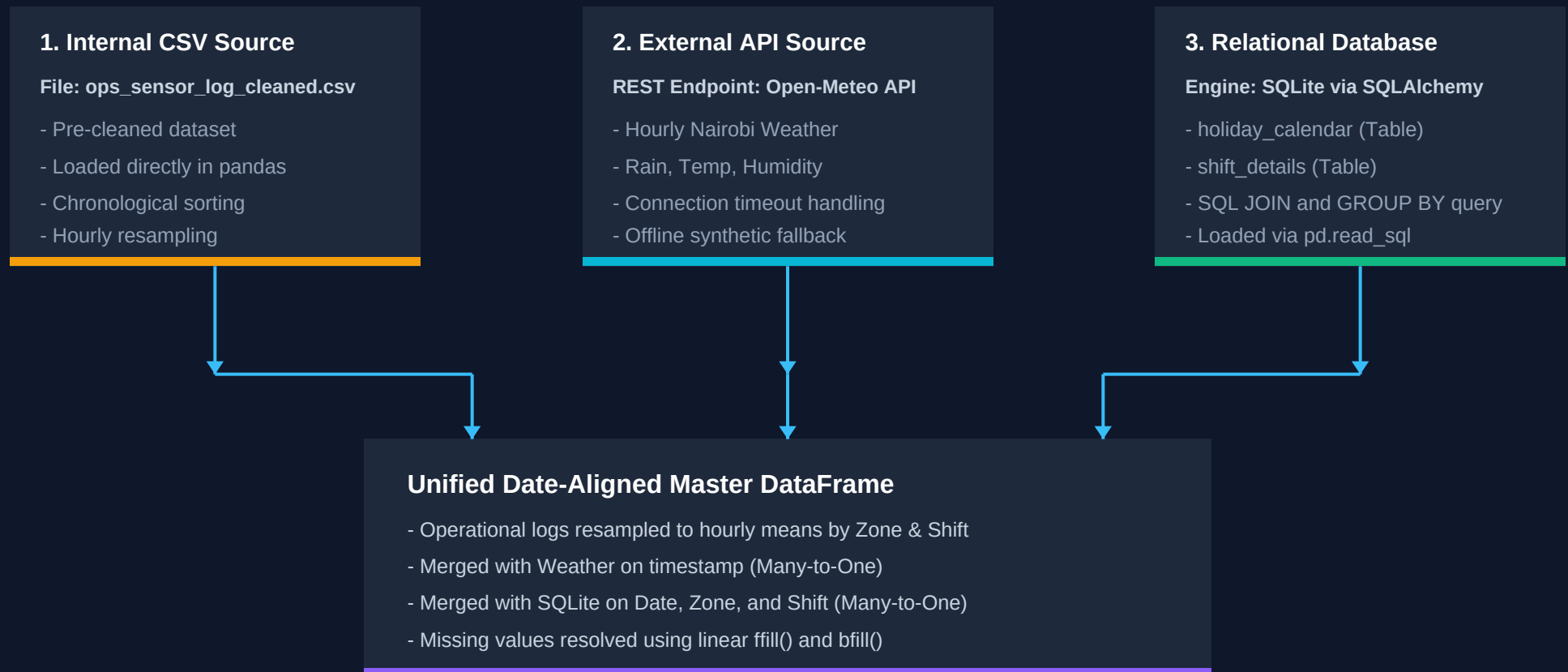
Key Operational Questions Investigated:

- Environmental Impact: Did heavy precipitation trigger pump line friction or viscosity drops?
- Workforce Governance: Did team sizes or supervisor experience on the holiday night shift affect throughput?
- Machine Stability: Did localized pressure limits (100 - 300 PSI) remain within stable thresholds?

The Data Integration Architecture

PIPELINE WORKFLOW: CSV + REST API + SQLITE DB

We developed a robust ETL pipeline merging three independent data sources into a standardized, date-aligned master hourly dataset.



KEY FINDINGS FROM MULTI-SOURCE ANALYSIS

1. The Weather Myth

Statistical Correlation $r = 0.00$

Our Pearson correlation matrix shows that rainfall has exactly $r = 0.00$ correlation with mean flow rates.

Insight:

Depot pumps and lines are physically shielded, meaning precipitation does NOT directly retard pipe hydraulics.

2. Governance & Staffing

Night Shift Capacity Cut by 50%

Night shifts operate with a planned team size of 4, compared to 8 on Morning shifts.

Insight:

On holidays and rainy nights, this capacity drop leads to delayed maintenance recovery, resulting in prolonged system throttling.

3. Actionable Strategy

Director Recommendations

Operational changes to make:

1. Weather-Adaptive Staffing:
Increase holiday night shifts from 4 to 6 when storm warnings are active.

2. Supervisor Reallocation:
Assign top performing leads (Munyao, Njoroge) to holiday night shifts during wet seasons.